

Insurance Lead Conversion & Marketing Performance Analysis

Project Overview

This project analyzes an insurance lead dataset to understand customer conversion, revenue generation, marketing campaign performance, and customer segmentation. The workflow included data cleaning using Python, exploratory data analysis (EDA), and the development of an interactive Power BI dashboard to support business decision-making.

Data Cleaning & Preparation

- Removed duplicate records.
- Converted date columns to datetime format.
- Converted premium values to numeric format.
- Standardized categorical values by removing extra spaces and inconsistent text formatting.
- Preserved booking-related missing values because they represent leads that did not convert into booked policies.
- Exported the cleaned dataset for Power BI dashboard development.

Business Findings

Lead Funnel Analysis

- 73,907 total leads and approximately 2,184 booked policies, resulting in an overall conversion rate of about 3%.
- Quote stage was not available in the dataset, therefore the funnel was analyzed as Lead -> Booking.

Premium & Revenue Analysis

- Average premium per booked policy is approximately ₹996.
- NIC Insurance generated the highest premium revenue.
- Comprehensive (Comp) plans contributed the highest share of revenue.

Marketing Performance

- UTM Medium and UTM Campaign performance were evaluated using bookings and revenue.
- A small number of campaigns generated the majority of bookings and premium revenue.
- Customer & Demographic Analysis
- Hero MotoCorp recorded the highest number of booked policies.
- New Delhi generated the highest number of bookings.
- The 76–150 CC segment showed the highest purchase rate.

Dashboard Summary

Three interactive Power BI dashboard pages were developed:

- Executive Dashboard
- Marketing Performance Dashboard
- Customer & Vehicle Insights Dashboard

Interactive filters allow analysis by insurer, plan type, campaign, marketing medium, and region.

Business Recommendations

- Increase investment in high-performing campaigns.
- Investigate campaigns with high lead volume but low conversion.
- Focus marketing on high-performing cities and vehicle segments.
- Strengthen partnerships with top-performing insurers.
- Continuously monitor conversion KPIs using the dashboard.

Bonus Case Study

Scenario: Booked insurance policies suddenly decrease despite an increase in incoming leads.

Investigation Approach

- Identify where the funnel is breaking.
- Compare campaign-wise conversion rates.
- Review pricing and insurer availability.
- Analyze customer segment changes.
- Check CRM, payment, and operational issues.
- Reallocate budget to high-converting campaigns and monitor KPIs regularly.